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Report on the PhD thesis of Zhanhong He entitled:
"Recovering Musical Dynamics for Expressive Automatic Music Transcription"

Introduction

Zhanhong He has provided a PhD thesis which consists of 154 pages, 127 of which make up the main body of the text which consists of seven chapters. The core methodological chapters are chapters 3-6, with the first two providing an introduction and background and the last chapter being the thesis conclusions. Overall the document is well written and clearly organised, with no comments on the overall chapter structure. In terms of citation style, I would suggest to use an author-year format which improves readability given a lack of space constraints. The use of language and the overall presentation of the text is excellent. In general the thesis is of high quality and was a pleasure to read.

Three works carried out by the candidate have been published as peer-reviewed papers at the CMMR 2025, SMC 2026, and IEEE ICASSP 2026 conferences respectively, and one work has been accepted for publication at the ISMIR 2026 conference. This shows good research productivity and consistency on publishing the outputs of Zhanhong's PhD research to the music computing and MIR communities. All associated publications were carried out by Zhanhong as lead author with supervisors as co-authors.

Thesis Evaluation

According to the six merit criteria set by the University of Western Australia:

- The thesis of Zhanhong He as a whole is a substantial and original contribution to knowledge in the fields of music information retrieval and music signal processing (the latter particularly for the contribution in chapter 6).
- As demonstrated by Chapter 2 of the thesis, but also throughout the thesis document, the candidate shows familiarity with, and understanding of, the relevant literature particularly on topics related to automatic music transcription and velocity prediction, and more broadly on topics related to music information retrieval and music signal processing (such as DDSP).
- The thesis provides a comprehensive study of the topic of automatically recovering musical dynamics from music audio and machine-readable symbolic music representations, perhaps at a level not achieved before in the field.
- The techniques adopted and proposed (primarily computational, related to machine learning and signal processing) are all appropriate to the subject matter of the thesis and are properly applied. In certain cases, as with most engineering works, there could be alternative techniques or minor suggestions on certain experimental settings; these will be detailed in the comments below.
- For each of the core methodological chapters (3-6), there are well-designed experiments and evaluations using appropriate existing or newly proposed evaluation metrics; results are suitably set out in tables and figures, and there is a good level of discussion of experimental results. As with most PhD theses including a computational evaluation component, there can be minor suggestions for presentation of results or choice of experimental setup; these are detailed in the comments below.
- The quality of English and general presentation are of a standard for doctoral thesis publication. The narrative in particular flows very well across and within chapters.

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Specific comments and questions

Below are specific comments on the thesis manuscript, as well as topics to be discussed during the viva.

Preamble: Submitted ISMIR 2026 paper can now be marked as accepted for publication.

Chapter 1:

- Figure 1.1 is captioned as a "general architecture of an AMT system", however step 2 of the figure only takes into account dynamics and velocity estimation, and not other aspects of music performance (e.g. instrument identities, expressive playing techniques).
- Section 1.1, final paragraph: the term "recollecting" is ambiguous, perhaps "collecting" is more appropriate?
- Section 1.5: how do you define weakly expressive and weakly supervised music data?
- Table 1.2: would a scenario of converting performance audio to MIDI velocity also apply, particularly for the work done in chapter 6?

Chapter 2:

- Figure 2.1: does there need to be always a correspondence between the Perceptual level and the MIDI velocity level? Could there be a direct link between the Perceptual level to the Score Dynamics level?
- Table 2.3 could benefit from inserting vertical lines to separate triplets of columns (1-3 vs 4-6 vs 7-9).
- Section 2.3.2: "MIDI [...] is attached to a note onset". Does that always hold true for the MIDI protocol?
- Figure 2.3: In subfigures (e) and (f), it is not clear what do the different colours signify.
- Section 2.5: The term "humanisation" is unclear – I would suggest to either define or replace it.
- Section 2.6.1 could also discuss other score dynamics symbols, including crescendo and diminuendo.
- Table 2.7: should an "audio" column be included for those datasets which include audio recordings? The upper triangular symbol is unclear. Does the MIDI column include both performance (unquantised) MIDI and reference (quantised) MIDI? What does the "43%" refer to? Many of the above can be clarified in the table caption.
- Section 2.8.3: It might benefit the background section if commonly used evaluation metrics for dynamics prediction are defined here.

Chapter 3:

- Section 3.1: This work is mentioned as the "most controlled" in the thesis; however predicting dynamics from a score in the absence of any dynamics markings or velocity info is a more unconstrained problem than dynamics prediction given input audio (where dynamics are present).
- Section 3.1 and beyond: to what extent should MIDI Velocity completion be style-specific or composer-specific?
- Section 3.2 and beyond considers the proposed work as "image-based". Although there are some related figures with varying colours included in this chapter, I do not see any strong argument for using this term – the methodology does not treat dynamics as a colour channel or similar.
- Section 3.3: The text and candidate could perhaps better justify the use of piano-roll as representation, as opposed to other MIDI-like representations (with lists of notes).
- Section 3.3: is this task about velocity recovery? Or velocity estimation?
- Section 3.4.1: could the input representation be in musical time as opposed to physical time? One might expect that velocity prediction in symbolic music might be more accurate using a beat-aligned representation.
- Section 3.6: The data used for experiments is performance MIDI, as opposed to reference (quantised) MIDI. Would that conflict with the main envisaged usecase which is on music composition?
- Section 3.7: Baseline models could be presented in a separate subsection with more detail.

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Chapter 4:

- Section 4.4: As part of the motivation for this work, it is stated that the proposed system isolates dynamics refinement from note transcription. Although practical, is this ecologically valid? Also, there is evidence that good velocity prediction can improve on note transcription performance, which would motivate linking both tasks and not isolating them.
- Section 4.5 and Figure 4.2: The use of piano-rolls for the proposed work could be further motivated; why not use event-based representations which are closer to MIDI?
- Section 4.5.3: Are both loss components balanced? Is there a need to weigh loss components?
- The different experimental setups as presented in Table 4.1 are not clearly defined in the main text of the thesis.

Chapter 5:

- Section 5.3: beyond the dynamics-related predictions made by the proposed system, did you attempt to combine these outputs with the outputs of a note-based AMT system as to have a complete AMT system?
- Section 5.3: It is not clear why there is a need to explictly model changepoints, and not predict dynamics markings at beat/downbeat levels.
- Figure 5.2: Are the beat and downbeat components not connected? The same applies to the loss function of equation 5.2.
- Section 5.6.1: is cross-dataset evaluation crucial for this specific work? Given that dynamics prediction can be to a large extent composer- and performer-dependent.

Chapter 6:

- Section 6.2: The author argues that this chapter is taking a "simpler route". However I would doubt that treating the problem using analysis-by-synthesis methods and DDSP is simpler than a supervised labelling effort. One might argue that the former is more appropriate and accurate, and the latter is less valid and time-consuming, however modelling simplicity might be difficult to argue for the proposed work.
- Section 6.4.1: What is the difference between V^{(0)} and \hat{V} ?
- Section 6.4.2: Can you comment on the faithfulness of the DDSP-Piano and DDSP-Guitar-Synth synthesizers?
- Section 6.4.3: Can you justify the use of a Huber distance as opposed to a more straightforward MAE loss?
- Figure 6.2: is the data in the right part of the figure rendered audio? If yes, this is not shown in the figure.
- Section 6.9.4: Could the SoundFont selection not be selected by ear but be automated instead?

Chapter 7:

- Would it be possible to integrate the developed models into AMT models for complete AMT? Would these improve on transcription performance?
- Section 7.3: A possible limitation of this work is that not all possible dynamics markings (e.g. crescendo) are invetigated; these could be mentioned as future work.

Overall recommendation

As demonstrated by this thesis, the research carried out by the candidate contributed new knowledge to the field of music information retrieval in terms of proposing new models for predicting or recovering musical dynamics in the context of automatic music transcription. This is a systematic and thorough work into the topic, with significant contributions to knowledge, and the proposed work was carried out in a systematic and rigorous manner. As can be seen from the comments above, I have certain minor queries or comments that can be further discussed during the viva. I therefore approve the PhD thesis of Zhanhong He to proceed to viva, and consider the written work to be excellent, at the level of top 25% PhD theses I have examined.

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Yours faithfully,



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